

MDRNet: Multi-Branch with Different Feature Representations Network for Motor Imagery Classification

Qiaoli Zhou^{1,2}, Yi Zhao¹, Shun Zhang³, Jiawen Song¹, Qiang Du², Li Ke^{2*}

¹ School of Computer Science, Shenyang Aerospace University, Shenyang 110136, Liaoning, China

² School of Electrical Engineering, Shenyang University of Technology, Shenyang 110870, Liaoning, China

³ Beijing Zhongke Sichuang Cloud Intelligent Technology Co., Ltd, 100083, Beijing, China

Abstract—A brain-computer interface (BCI) offers an innovative solution for facilitating communication and control in individuals with paralysis. BCI reflects brain activity by decoding electroencephalogram (EEG) signals. Despite numerous techniques for classifying motor imagery (MI) EEG signals, challenges such as low signal-to-noise ratio (SNR), nonlinearity, non-stationarity, and feature information loss continue to persist. This paper proposes an innovative Multi-Branch with Different Feature Representations Network (MDRNet) to address these issues. Firstly, we introduce the Temporal Gramian Angular Summation Field (T-GASF) method, which enhances signal features with temporal information, thereby improving SNR. Secondly, we present a hybrid attention mechanism based on empirical mode decomposition (EMD), combining channel attention and intrinsic mode function (IMF) attention to effectively address signal nonlinearity and non-stationarity. Lastly, we develop a U-net branch with T-GASF input, a Channel-IMF Hybrid Attention (CIHA) branch based on EMD-decomposed signals, and a Transformer Encoder-CNN branch. These branches ensure comprehensive extraction of critical EEG features. Experimental findings from the BCI Competition IV 2a and 2b datasets indicate that MDRNet attained classification accuracies of 95.71% and 90.28%, respectively, markedly outperforming current methods.

Index Terms—Electroencephalogram, Motor Imagery, Deep Learning, Attention

I. INTRODUCTION

A brain-computer interface (BCI) establishes a communication link between the brain and external devices. It interprets the user's intentions by identifying specific tasks associated with neuronal activity and converting them into computer commands [1]. Among the various techniques used to capture neural activity, electroencephalography (EEG) is widely favored due to its non-invasive nature and excellent temporal resolution. Motor imagery (MI) represents a significant field of study within EEG research. MI refers to EEG signals generated by the sensorimotor cortex when an individual mentally simulates moving a body part without physically executing the movement [2]. Imagined tasks in MI can involve different body parts, such as the left hand, right hand, and foot [3].

In traditional MI processing methodologies, feature representation and feature extraction are the most crucial aspects. The feature representation of EEG is primarily analysed across four domains: time domain [4], frequency domain [5], time-frequency domain [6], and spatial domain [7]. Specifically, Rahman et al. [8] developed a frequency-time localization feature extraction technique, conducting analyses of the signal in both the frequency and time domains. Hsu et al. [9] proposed a novel method, the wavelet-based temporal-spectral-attention correlation coefficient (WTS-CC), which enables comprehensive signal analysis across the time, frequency, and spatial domains.

*Corresponding author: Li Ke (keli@sut.edu.cn)

This research was funded by the National Nature Science Foundation of China, grant number 52077143.

After feature representation, effective feature extraction is critical for subsequent classification. Deep learning (DL) algorithms can autonomously extract meaningful features from EEG data and perform high-accuracy classification. For instance, Musallam et al. [10] proposed TCNet-Fusion, which learns time-domain information through time-domain convolution. However, EEG signals contain vast information, and relying on a single domain can reduce model robustness. To address this, Jia et al. [11] introduced TSF-STAN, which captures both temporal and spatial data, but single-branch networks may still inadequately capture multi-domain information.

To solve this, multi-branch models have gained attention. Huang et al. [12] proposed a dual-stream CNN (DCNN) that combines time- and frequency-domain data for classification. Wang et al. [13] introduced MSFNet, optimizing EEG classification by integrating information across temporal, spatial, and frequency domains. Thus, multi-branch networks offer significant advantages in learning the multi-dimensional features of EEG data.

When processing MI-EEG signals, they are typically affected by various physiological and non-physiological noises, such as electromyographic noise, power line interference, and environmental noise. The presence of these disturbances results in a diminished SNR in EEG data, which consequently affects the precision of subsequent tasks. Fortunately, feature representation can mitigate this issue. Furthermore, MI-EEG signals contain a significant amount of nonlinear and non-stationary information, which can degrade model performance. Lastly, ensuring the integrity of feature representation while extracting effective features from EEG signals presents another significant challenge.

The innovations introduced in addressing EEG signal processing issues are as follows:

- We introduce the Temporal Gramian Angular Summation Field (T-GASF) method, an improvement based on GASF. This method integrates temporal information to improve the representation of temporal features in signals, thereby enhancing the SNR.
- We propose a hybrid attention mechanism called Channel-IMF Hybrid Attention (CIHA), based on intrinsic mode functions (IMF) obtained through empirical mode decomposition (EMD). This method combines Channel Attention and IMF Attention to more effectively handle the nonlinear and non-stationary characteristics of EEG data.
- We introduce an Adaptive Multi-head Attention mechanism in the IMF Attention module to dynamically assign attention weights to different IMFs. This mechanism allows the model to emphasize informative IMFs and reduce the impact of noisy IMFs, further enhancing the feature extraction process for nonlinear and non-stationary EEG data.

- We developed a Multi-Branch with Different Feature Representations Network (MDRNet). It includes three branches: a U-net branch with T-GASF input, a CIHA branch with EMD-decomposed signal input, and a Transformer Encoder-CNN branch. This design ensures comprehensive and accurate extraction of important features in EEG signals.

II. METHODOLOGY

A. Proposed Architecture

This paper introduces a method called MDRNet, as shown in Fig. 1. MDRNet comprises three main components: feature representation, feature extraction, and classification, specifically designed for analyzing EEG signals. Initially, in the feature representation phase, the EEG data undergoes processing by an autoencoder in order to minimize the error in reconstruction. Subsequently, the EEG signals undergo T-GASF transformation and EMD processing. The EEG data, after feature representation, is then fed into three feature extraction modules: the Temporal Image Feature Extraction Module (TFM), the Multi-Dimensional Channel Relationship Extraction Module (MDCM), and the EEG Signal Temporal Relationship Extraction Module (ETM). Finally, in the classification, the feature vectors extracted by each module are concatenated, and the aggregated features are classified using a Multi-Layer Perceptron (MLP).

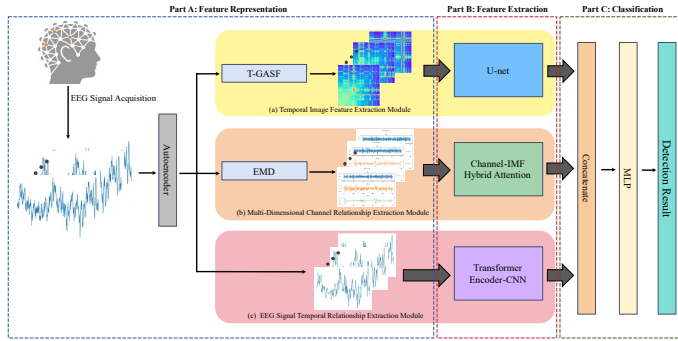


Fig. 1. The structure of the Multi-Branch with Different Feature Representations Network (MDRNet).

B. Temporal Image Feature Extraction Module

In the preprocessing of EEG signals, time-encoding techniques are crucial, especially in extracting temporal correlations from the signals. Among these techniques, Gramian Angular Field (GAF) [14] is a highly efficient method. By converting EEG signals into temporally correlated feature representations, GAF can capture the temporal dependencies within the signals. This allows for a more accurate reflection of the dynamic characteristics of brain activity.

Consider a time series $X = \{x_1, x_2, \dots, x_L\}$ of length L normalized to the range $[-1, 1]$. The transformation to polar coordinates involves the following steps:

- Transform the data into polar coordinates:

$$\phi_i = \arccos(x_i) \quad (1)$$

where x_i denotes the value of the time series at timestamp t_i , and ϕ_i represents the corresponding angular value.

- Regularize the timestamps:

$$r_i = \frac{t_i}{N} \quad (2)$$

where N acts as a constant factor to regularize the span of the coordinate system, ensuring that the timestamps are appropriately scaled.

Through this coordinate transformation, the GASF can be constructed, which is defined as:

$$GASF = \begin{bmatrix} \cos(\phi_1 + \phi_1) & \cdots & \cos(\phi_1 + \phi_n) \\ \cos(\phi_2 + \phi_1) & \cdots & \cos(\phi_2 + \phi_n) \\ \vdots & \ddots & \vdots \\ \cos(\phi_n + \phi_1) & \cdots & \cos(\phi_n + \phi_n) \end{bmatrix} \quad (3)$$

In order to convert the implicit dimension of time into an explicit one, we introduce a temporal factor r_i into the formula to enhance the Gramian Angular Summation Field (GASF). The T-GASF can better capture the temporal variations of the data, and it is defined as follows:

$$T-GASF = \begin{bmatrix} r_1 \cos(\phi_1 + \phi_1) & \cdots & r_n \cos(\phi_1 + \phi_n) \\ r_2 \cos(\phi_2 + \phi_1) & \cdots & r_n \cos(\phi_2 + \phi_n) \\ \vdots & \ddots & \vdots \\ r_n \cos(\phi_n + \phi_1) & \cdots & r_n \cos(\phi_n + \phi_n) \end{bmatrix} \quad (4)$$

U-net [15] is a CNN architecture initially developed for biomedical image segmentation. Its ability to capture both fine-grained details and broader regions makes it an ideal choice for extracting features from T-GASF. Therefore, we utilize U-net to extract temporal information from T-GASF.

C. Multi-Dimensional Channel Relationship Extraction Module

In this section, we introduce the MDCM, designed to capture and analyze the complex relationships within EEG signals. This module employs EMD techniques to decompose the signals and apply attention mechanisms for comprehensive feature extraction.

1) *EMD*: EMD [16] is an effective method for decomposing a signal into a series of intrinsic oscillatory components, known as IMFs, allowing for the analysis of non-stationary and non-linear signals. After decomposition, the IMFs effectively characterize the original signal, making the analysis of complex signals more straightforward. For instance, in Fig. 2 (a), the original EEG signal is depicted, showing typical non-linear and non-stationary properties. As shown in Fig. 2, the EMD decomposition effectively extracts the distinct oscillatory components of the signal. Fig. 2 (a) illustrates the original EEG signal with its complex fluctuations, whereas Fig. 2 (b) shows the first three IMFs extracted through EMD, presenting a more regular and stationary pattern that facilitates subsequent signal processing and analysis.

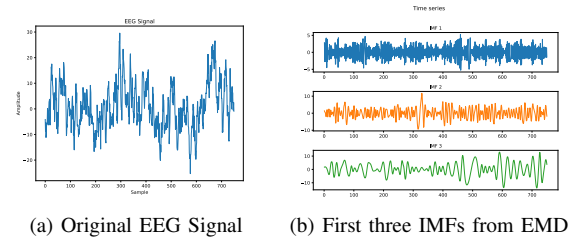


Fig. 2. Original EEG signal and corresponding IMFs derived from the EMD algorithm.

2) *Channel-IMF Hybrid Attention*: This paper introduces CIHA, a novel attention mechanism designed to enhance deep feature extraction from multi-channel EEG data. As shown in Fig. 3, CIHA consists of two main components: Channel Attention (Part A) and IMF Attention (Part B).

Part A is designed to extract inter-channel correlations by processing EMD-decomposed EEG signals. Specifically, given an input signal $X \in \mathbb{R}^{C \times (I \times T)}$, where C represents the number of channels, I is Number of IMFs and T is the time length, the multi-head attention mechanism is applied to capture diverse features across channels. The multi-head attention computes attention for each head h_i as follows:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (5)$$

where Q , K , and V represent the query, key, and value matrices, respectively, and d_k is the dimension of the key. After applying multi-head attention, the features are further refined using convolutional layers with Leaky ReLU activation and Batch Normalization. Finally, a fully connected (FC) layer is used to unify dimensions.

Part B focuses on extracting correlations between IMFs, incorporating an Adaptive Multi-head Attention(AMA) mechanism. Given the variability of information in different IMFs, some IMFs may carry more useful signal content, while others may contain noise. Therefore, we propose an adaptive attention mechanism that dynamically assigns weights to different IMFs, allowing the model to emphasize IMFs with richer information and downplay those dominated by noise.

Let the input IMFs be denoted as $I = \{IMF_1, IMF_2, \dots, IMF_K\} \in \mathbb{R}^{K \times T}$, where K is the number of IMFs. The adaptive multi-head attention computes the attention score for each IMF IMF_i as:

$$\alpha_i = \frac{\exp(e_i)}{\sum_{j=1}^K \exp(e_j)} \quad (6)$$

where e_i is a learnable parameter representing the importance of IMF i . The adaptive attention weights α_i are then applied to the corresponding attention heads, modulating their contributions:

$$\text{Adaptive Attention}(Q, K, V) = \sum_{i=1}^K \alpha_i \cdot \text{Attention}(Q_i, K_i, V_i) \quad (7)$$

This ensures that more informative IMFs receive higher attention weights, improving the signal extraction process. After applying adaptive multi-head attention, the IMF features are refined in a similar manner as in Part A, using convolutional layers and Batch Normalization. These refined IMF features are then unified through an FC layer.

By combining Channel Attention and IMF Attention with adaptive weighting, CIHA effectively captures both inter-channel and inter-IMF correlations. The adaptive mechanism in Part B ensures that noise-dominated IMFs contribute less to the final representation, while informative IMFs are emphasized. This comprehensive feature extraction approach enhances the overall model performance by balancing signal and noise contributions.

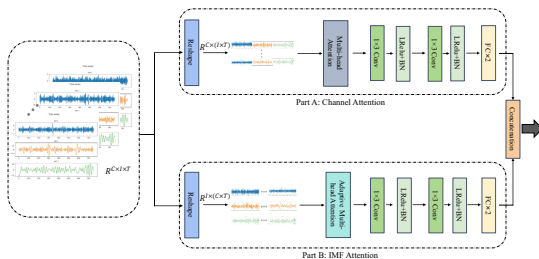


Fig. 3. The framework of CIHA.

D. EEG Signal Temporal Relationship Extraction Module

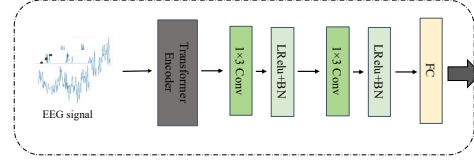


Fig. 4. EEG Signal Processing Architecture Based on Transformer Encoder and Convolutional Neural Network.

To address the information loss during feature extraction, we introduce a module, as illustrated in Fig. 4, that enhances feature learning from EEG data by combining the strengths of both global and local feature extraction techniques.

III. EXPERIMENTAL SETUP

A. Datasets and Experimental Setups

This paper evaluated MDRNet using two mainstream MI datasets, BCI-IV-2a [17] and BCI-IV-2b [18]. Table I presents the main information of these two datasets.

TABLE I
MAIN INFORMATIONS OF TWO DATASETS.

Datasets	Subjects	Channels	Sample rates(Hz)	Classes
BCI-IV-2a	9	22	250	4
BCI-IV-2b	9	3	250	2

For the two datasets, the subject-independent method's performance was assessed. Table II outlines a set of key hyperparameters. In this paper, accuracy and kappa value were employed to assess the performance of the proposed method. Additionally, five-fold cross-validation was utilized to ensure the robustness of the results.

TABLE II
MAIN HYPERPARAMETERS FOR TRAINING MDRNET ARCHITECTURE.

Datasets	BCI-IV-2a	BCI-IV-2b
Epoch	200	200
Batchsize	32	64
Learning rate	0.0001	0.0001

B. Results on BCI-IV-2a and BCI-IV-2b Datasets

We conducted experiments on both the BCI-IV-2a and BCI-IV-2b datasets, with outcomes compared against several state-of-the-art approaches. As shown in Tables III, MDRNet demonstrated more consistent and superior performance across all subjects in both datasets. On the BCI-IV-2a dataset, MDRNet achieved an average accuracy of 95.71%, a standard deviation of 2.09, and a kappa value of 0.89, highlighting a significant improvement over existing methods. Similarly, on the BCI-IV-2b dataset, MDRNet outperformed other approaches with an average accuracy of 90.28%, a standard deviation of 6.20, and a kappa value of 0.807, showcasing its stability and accuracy. These results underscore the significant advantages of MDRNet in EEG signal classification tasks, particularly in enhancing accuracy and consistency through its integration of temporally enhanced GAF, the EMD-based hybrid attention mechanism, and the three-branch hybrid network, which together contribute to its outstanding performance and robustness.

TABLE III
A DETAILED COMPARISON OF PERFORMANCE (ACCURACY(%) AND KAPPA) BETWEEN DIFFERENT METHODS AND ACROSS VARIOUS YEARS ON THE BCI-IV-2A AND BCI-IV-2B DATASETS.

Datasets	Methods	Years	S01	S02	S03	S04	S05	S06	S07	S08	S09	Average	Std	Kappa
BCI-IV-2a	MSFFCNN-TL [19]	2022	93.64	93.37	96.76	92.17	95.59	91.07	94.56	97.14	92.36	94.06	-	0.88
	TCACNet [20]	2022	91.3	65.2	93.9	84.3	90.4	75.7	98.3	90.4	91.3	86.8	10.3	-
	WMBEENet [21]	2023	92.01	68.06	97.22	76.74	78.47	70.14	94.44	91.67	88.54	84.14	-	0.79
	3D CNN [22]	2023	97.22	84.1	98.37	92.33	92.71	81.98	99.34	95.38	96.15	93.06	-	-
	iWSGL-CSP [23]	2024	92.96	65.19	97.41	78.33	75.19	77.04	88.15	97.41	94.78	85.13	11.56	0.7
	MDRNet(ours)		98.18	91.89	97.26	96.93	95.47	93.10	97.45	95.14	96.00	95.71	2.09	0.89
BCI-IV-2b	DTCWT-CSP [24]	2022	85.6	66.3	63.7	99.4	91.3	79.2	86.9	94.4	89.4	84.02	12.2	0.68
	FBRTS [25]	2022	82.4	75.2	86.9	95.2	89.7	80.2	90.5	91.2	91.1	86.9	6.03	0.70
	WTS-CC [9]	2023	81.11	78.14	76.88	91.11	71.66	91.51	86.25	74.61	84.79	81.78	7.1	-
	CMO-CNN [26]	2023	86.38	77.27	79.44	97.29	94.25	88.33	89.79	86.31	85.69	87.19	-	-
	WMB-EEGNet [21]	2023	84.69	73.93	84.38	97.81	100	90.31	94.06	95.00	91.88	90.23	8.09	0.80
	TSFF-Net [27]	2024	83.8	66.8	67.8	98.4	95.6	90.0	86.2	93.8	95.0	86.4	11.8	0.728
	MDRNet(ours)		87.82	81.03	81.73	97.39	98.26	87.82	93.96	93.10	91.37	90.28	6.20	0.807

C. Visualization of EEG Signals in MI Tasks with T-GASF

Fig. 5 shows the EEG signals from different channels in MI tasks, along with their corresponding GASF and T-GASF images. The GASF images capture temporal dependencies by converting EEG signals into temporally correlated features. In contrast, the T-GASF images enhance the GASF by incorporating a time factor, providing a more comprehensive depiction of the temporal dependencies and internal temporal variations of the EEG signals. In the T-GASF images, the diagonal running from the top left to the bottom right clearly reflects the trend of temporal changes. This improvement allows T-GASF images to not only retain the temporal correlations of the signals but also reveal the variation patterns at different time points, thereby providing richer information for further feature extraction and classification.

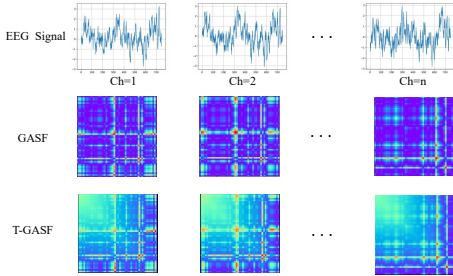


Fig. 5. EEG signals from different channels in MI tasks, along with their corresponding GASF images and T-GASF images.

D. Ablation Studies

We proposed the MDRNet architecture, which comprises two key components: the T-GASF method for enhanced time series data processing and the CIHA mechanism. To validate its effectiveness, we conducted both module ablation and branch ablation experiments on the BCI-IV-2b dataset. The outcomes of these experiments are visually presented in Fig. 6.

We initially assessed the impact of using the original GAF method, observing a 0.5% decrease in accuracy when switching to GASF, as illustrated in Fig. 6 (a), demonstrating the effectiveness of T-GASF in capturing temporal dependencies. Further analysis using only the self-attention mechanism resulted in an additional 1.27% drop in accuracy, underscoring the critical role of the CIHA mechanism in feature extraction. When comparing the self-attention model with the hybrid attention model, CIHA demonstrated superior relational information extraction, leading to enhanced performance. Additionally, the use of

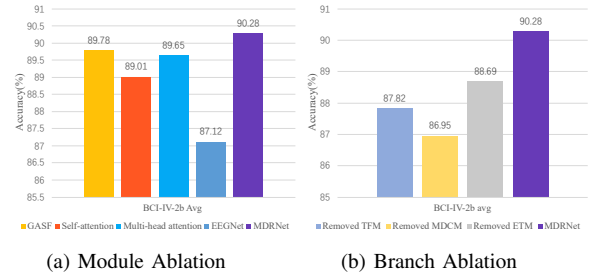


Fig. 6. Ablation study results of MDRNet on the BCI-IV-2b dataset.

standard Multi-head Attention in place of the adaptive mechanism in the IMF Attention module caused a 0.63% decrease in accuracy, which proves that different IMFs contain varying amounts of useful information, with some IMFs being more informative and others primarily containing noise. Lastly, replacing MDRNet with EEGNet [28] resulted in a significant 3.16% decrease in accuracy, confirming that the MDRNet architecture is more effective for specialized feature learning.

Additionally, branch ablation experiments were conducted on MDRNet. As shown in Fig. 6 (b), removing the TFM led to a 2.46% decrease in accuracy, indicating the crucial role of the T-GASF branch in temporal feature extraction. The accuracy dropped by 3.33% upon removing the CIHA branch, underscoring its importance in handling nonlinear and non-stationary features. Finally, eliminating the ETM led to a 1.59% reduction in accuracy, demonstrating its role in preserving comprehensive feature representation. These findings collectively highlight the significant contributions of the TFM, MDCM, and ETM to the high performance of MDRNet.

IV. CONCLUSION

This paper presents MDRNet, a novel architecture for processing EEG signals that integrates T-GASF and CIHA modules. Our approach effectively addresses the challenges of low SNR, nonlinearity, non-stationarity, and feature information loss in EEG signal classification. The key contributions of our work include the improved GASF with temporal information, which enhances signal features and improves SNR; the hybrid attention mechanism combining channel and IMF-based self-attention, which effectively addresses signal nonlinearity and non-stationarity; and a three-branch network that ensures comprehensive extraction of critical EEG features. These innovations collectively enhance the effectiveness of MDRNet in EEG signal classification tasks, as demonstrated by our experimental results on two datasets.

REFERENCES

- [1] B. He, H. Yuan, J. Meng, and S. Gao, "Brain-computer interfaces," *Neural engineering*, pp. 131–183, 2020.
- [2] Y. R. Tabar and U. Halici, "A novel deep learning approach for classification of eeg motor imagery signals," *Journal of neural engineering*, vol. 14, no. 1, p. 016003, 2016.
- [3] Y. Han, B. Wang, J. Luo, L. Li, and X. Li, "A classification method for eeg motor imagery signals based on parallel convolutional neural network," *Biomedical Signal Processing and Control*, vol. 71, p. 103190, 2022.
- [4] S. S. Gupta and R. R. Manthalkar, "Detection of motor activity in visual cognitive task using autoregressive modelling and deep recurrent network," in *Pattern Recognition and Data Analysis with Applications*. Springer, 2022, pp. 371–381.
- [5] A. C. Subrata, M. A. Riyadi, and T. Prakoso, "Eeg-based bmi using multi-class motor imagery for bionic arm," in *2020 3rd International Conference on Mechanical, Electronics, Computer, and Industrial Technology (MECnIT)*. IEEE, 2020, pp. 255–260.
- [6] J. Jin, T. Qu, R. Xu, X. Wang, and A. Cichocki, "Motor imagery eeg classification based on riemannian sparse optimization and dempster-shafer fusion of multi-time-frequency patterns," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 31, pp. 58–67, 2022.
- [7] X. Geng, S. Xue, P. Yu, D. Li, M. Yue, X. Zhang, and L. Wang, "A fusion algorithm for eeg signal processing based on motor imagery brain-computer interface," *Wireless Communications and Mobile Computing*, vol. 2022, pp. 1–14, 2022.
- [8] M. Rahman and H. T. Shuvo, "An improved mi recognition by localising feature extraction in both frequency and time domains," *Computer Methods in Biomechanics and Biomedical Engineering: Imaging & Visualization*, vol. 11, no. 5, pp. 1818–1830, 2023.
- [9] W.-Y. Hsu and Y.-W. Cheng, "Eeg-channel-temporal-spectral-attention correlation for motor imagery eeg classification," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 31, pp. 1659–1669, 2023.
- [10] Y. K. Musallam, N. I. AlFassam, G. Muhammad, S. U. Amin, M. Al-sulaiman, W. Abdul, H. Altaheri, M. A. Bencherif, and M. Algabri, "Electroencephalography-based motor imagery classification using temporal convolutional network fusion," *Biomedical Signal Processing and Control*, vol. 69, p. 102826, 2021.
- [11] X. Jia, Y. Song, L. Yang, and L. Xie, "Joint spatial and temporal features extraction for multi-classification of motor imagery eeg," *Biomedical Signal Processing and Control*, vol. 71, p. 103247, 2022.
- [12] E. Huang, X. Zheng, Y. Fang, and Z. Zhang, "Classification of motor imagery eeg based on time-domain and frequency-domain dual-stream convolutional neural network," *IRBM*, vol. 43, no. 2, pp. 107–113, 2022.
- [13] C. Wang, Y. Wu, C. Wang, Y. Ren, J. Shen, T. Pang, C. S. Chan, W. Ren, and Y. Yu, "Msfnet: A multi-scale space-time frequency fusion network for motor imagery eeg classification," *IEEE Access*, 2024.
- [14] Z. Wang and T. Oates, "Imaging time-series to improve classification and imputation," *arXiv preprint arXiv:1506.00327*, 2015.
- [15] O. Ronneberger, P. Fischer, and T. Brox, "U-net: Convolutional networks for biomedical image segmentation," in *Medical image computing and computer-assisted intervention—MICCAI 2015: 18th international conference, Munich, Germany, October 5–9, 2015, proceedings, part III 18*. Springer, 2015, pp. 234–241.
- [16] N. E. Huang, Z. Shen, S. R. Long, M. C. Wu, H. H. Shih, Q. Zheng, N.-C. Yen, C. C. Tung, and H. H. Liu, "The empirical mode decomposition and the hilbert spectrum for nonlinear and non-stationary time series analysis," *Proceedings of the Royal Society of London. Series A: mathematical, physical and engineering sciences*, vol. 454, no. 1971, pp. 903–995, 1998.
- [17] C. Brunner, R. Leeb, G. Müller-Putz, A. Schlögl, and G. Pfurtscheller, "Bci competition 2008–graz data set a," *Institute for Knowledge Discovery (Laboratory of Brain-Computer Interfaces), Graz University of Technology*, vol. 16, pp. 1–6, 2008.
- [18] R. Leeb, C. Brunner, G. Müller-Putz, A. Schlögl, and G. Pfurtscheller, "Bci competition 2008–graz data set b," *Graz University of Technology, Austria*, vol. 16, pp. 1–6, 2008.
- [19] A. M. Roy, "Adaptive transfer learning-based multiscale feature fused deep convolutional neural network for eeg mi multiclassification in brain-computer interface," *Engineering Applications of Artificial Intelligence*, vol. 116, p. 105347, 2022.
- [20] X. Liu, R. Shi, Q. Hui, S. Xu, S. Wang, R. Na, Y. Sun, W. Ding, D. Zheng, and X. Chen, "Tcacnet: Temporal and channel attention convolutional network for motor imagery classification of eeg-based bci," *Information Processing & Management*, vol. 59, no. 5, p. 103001, 2022.
- [21] H. Wang, J. Jiang, J. Q. Gan, and H. Wang, "Motor imagery eeg classification based on a weighted multi-branch structure suitable for multisubject data," *IEEE Transactions on Biomedical Engineering*, 2023.
- [22] X. Liu, K. Wang, F. Liu, W. Zhao, and J. Liu, "3d convolution neural network with multiscale spatial and temporal cues for motor imagery eeg classification," *Cognitive Neurodynamics*, vol. 17, no. 5, pp. 1357–1380, 2023.
- [23] B. Lu, F. Wang, S. Wang, J. Chen, G. Wen, and R. Fu, "Improvement of motor imagery electroencephalogram decoding by iterative weighted sparse-group lasso," *Expert Systems with Applications*, vol. 238, p. 122286, 2024.
- [24] N. Malan and S. Sharma, "Motor imagery eeg spectral-spatial feature optimization using dual-tree complex wavelet and neighbourhood component analysis, irbm, 43 (2022), 198–209," *Crossref, ISI*.
- [25] H. Fang, J. Jin, I. Daly, and X. Wang, "Feature extraction method based on filter banks and riemannian tangent space in motor-imagery bci," *IEEE journal of biomedical and health informatics*, vol. 26, no. 6, pp. 2504–2514, 2022.
- [26] X. Liu, S. Xiong, X. Wang, T. Liang, H. Wang, and X. Liu, "A compact multi-branch 1d convolutional neural network for eeg-based motor imagery classification," *Biomedical Signal Processing and Control*, vol. 81, p. 104456, 2023.
- [27] Z. Miao and M. Zhao, "Time-space-frequency feature fusion for 3-channel motor imagery classification," *Biomedical Signal Processing and Control*, vol. 90, p. 105867, 2024.
- [28] V. J. Lawhern, A. J. Solon, N. R. Waytowich, S. M. Gordon, C. P. Hung, and B. J. Lance, "Eegnet: a compact convolutional neural network for eeg-based brain-computer interfaces," *Journal of neural engineering*, vol. 15, no. 5, p. 056013, 2018.